
Analog IC Design

Lecture 07 Cascode Amplifiers

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Outline

- ☐ Recapping previous key results
- ☐ Cascode amplifier
- ☐ Telescopic vs folded cascode
- ☐ Super cascode

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MOSFET in Saturation

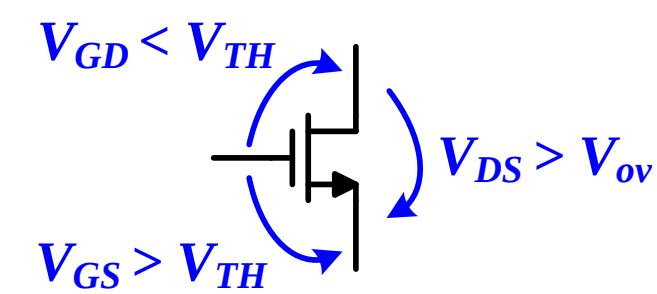
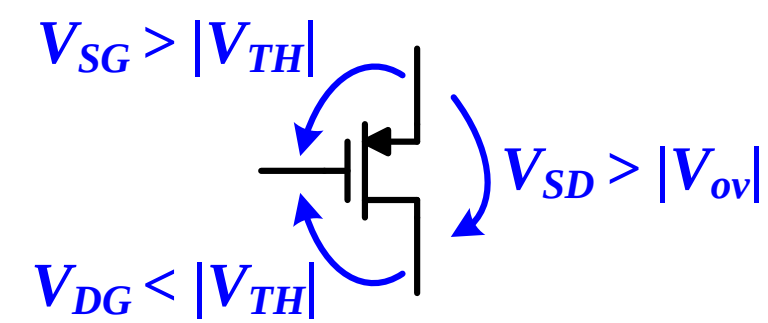
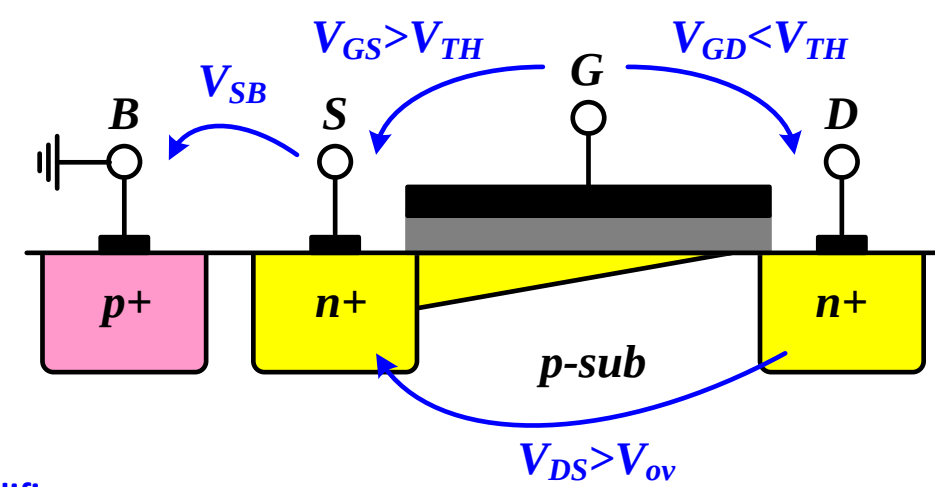
- ❑ The channel is pinched off if the difference between the gate and drain voltages is not sufficient to create an inversion layer

$$V_{GD} \leq V_{TH} \quad OR \quad V_{DS} \geq V_{ov}$$

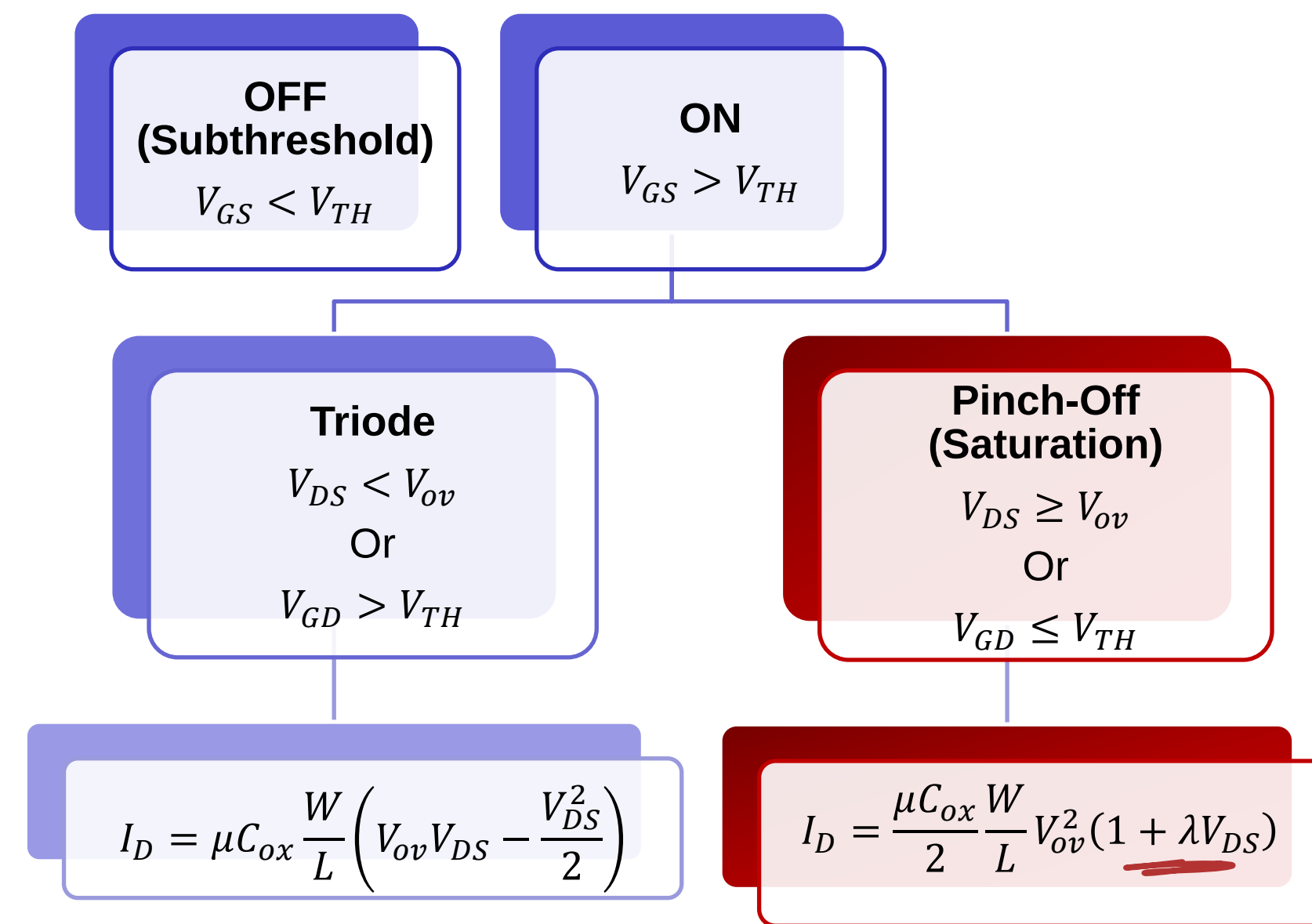
- ❑ Square-law (long channel MOS)

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot V_{ov}^2 (1 + \lambda V_{DS})$$

$$V_{SB} \uparrow \Rightarrow V_{TH} \uparrow$$



Regions of Operation Summary

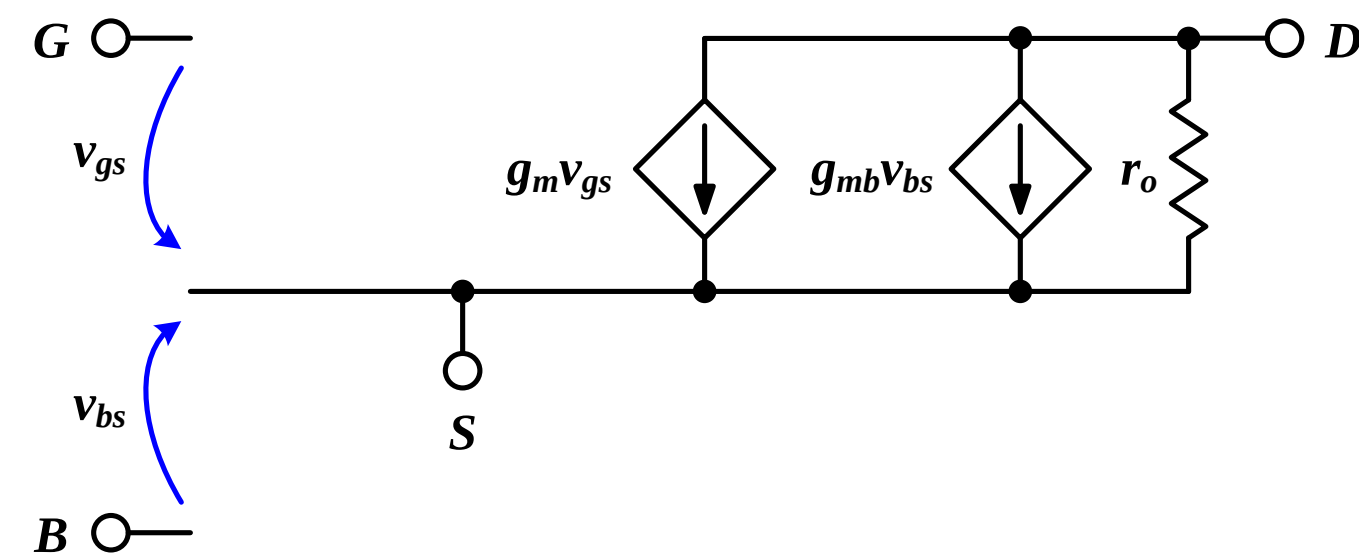


Low-Frequency Small-Signal Model

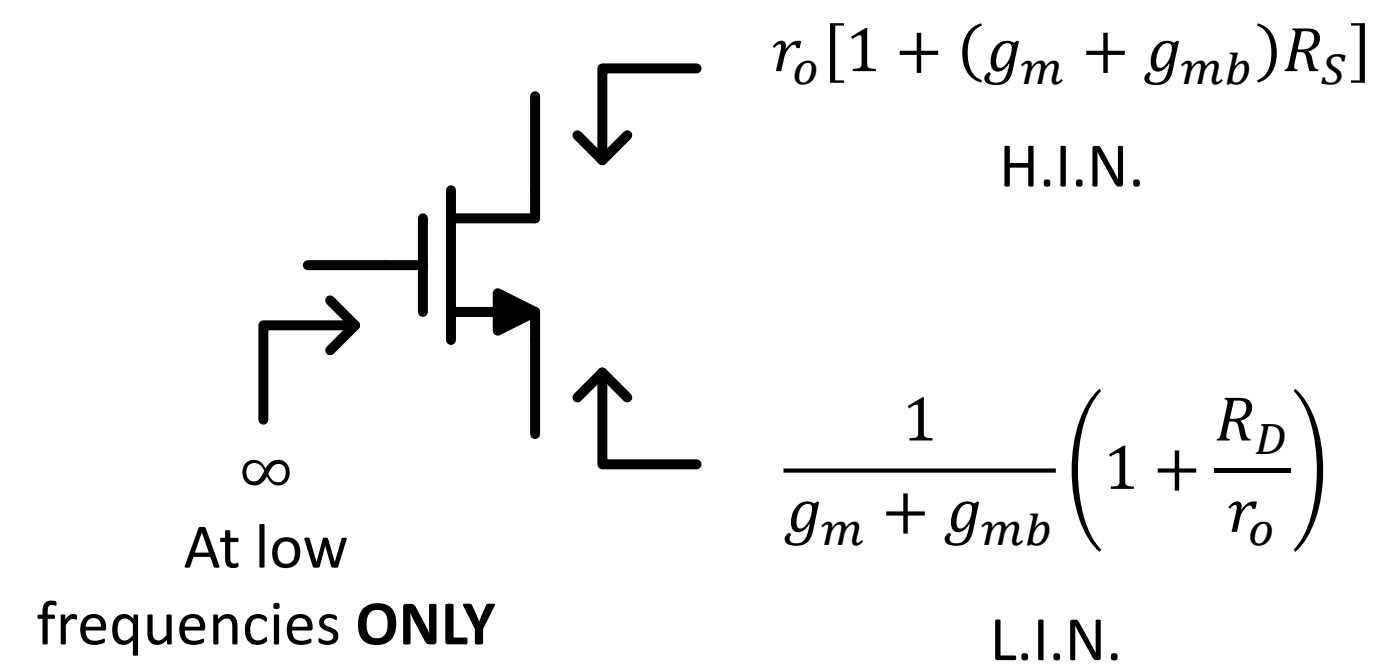
$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu C_{ox} \frac{W}{L} V_{ov} = \sqrt{\mu C_{ox} \frac{W}{L} \cdot 2I_D} = \frac{2I_D}{V_{ov}}$$

$$g_{mb} = \eta g_m \quad \eta \approx 0.1 - 0.25$$

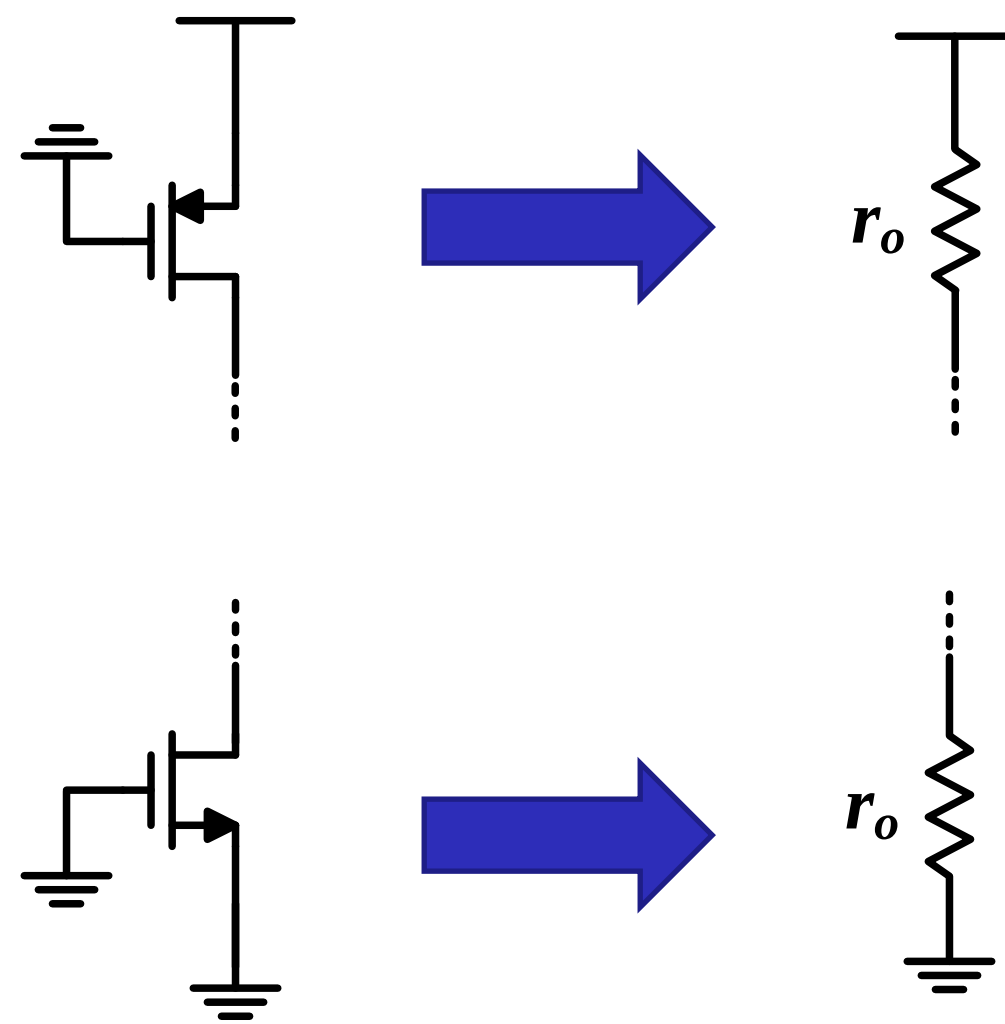
$$r_o = \frac{1}{\partial I_D / \partial V_{DS}} = \frac{V_A}{I_D} = \frac{1}{\lambda I_D} \quad V_A \propto L \leftrightarrow \lambda \propto \frac{1}{L} \quad V_{DS} \uparrow V_A \uparrow$$



Rin/out Shortcuts Summary

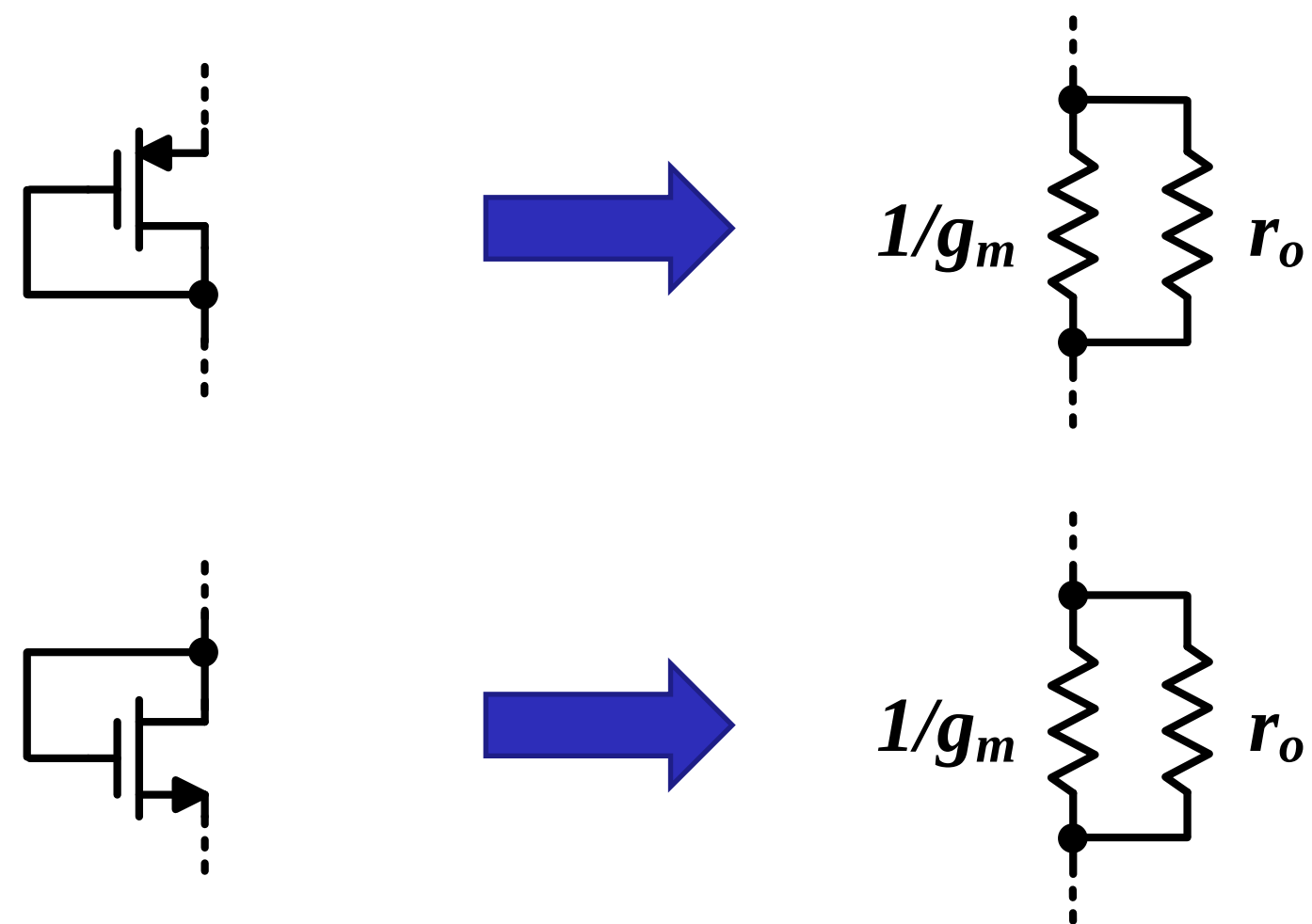


Active Load (Source OFF)



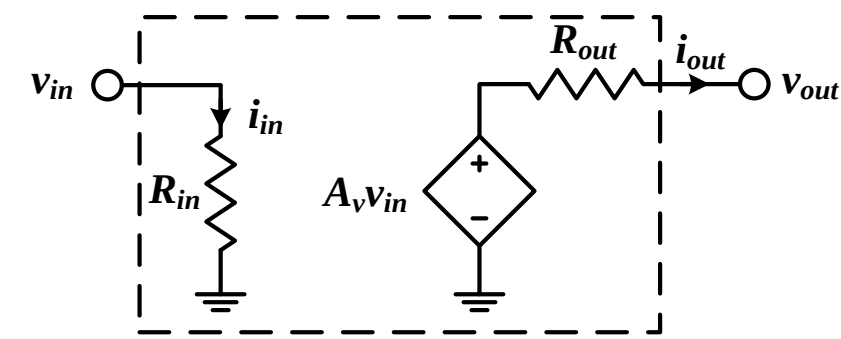
Diode Connected (Source Absorption)

- ❑ Always in saturation ($V_{DS} = V_{GS} > V_{ov}$)
- ❑ Body effect: $g_m \rightarrow g_m + g_{mb}$ (if G is ac gnd)



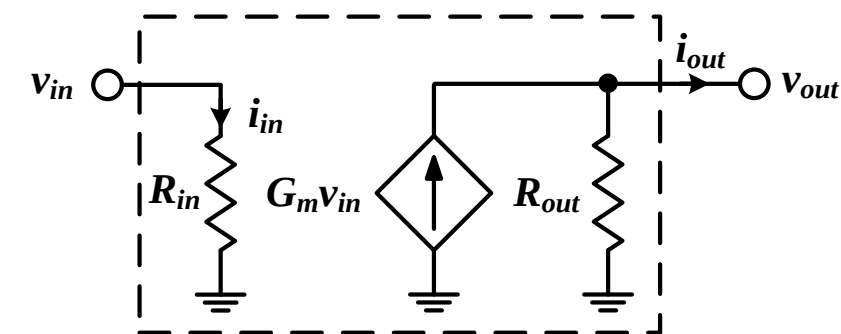
Why GmRout?

$$\begin{aligned} R_{in} &= v_{in}/i_{in} \\ R_{out} &= v_x/i_x @ v_{in} = 0 \\ G_m &= i_{out,sc}/v_{in} \\ A_v &= G_m R_{out} \\ A_i &= G_m R_{in} \end{aligned}$$

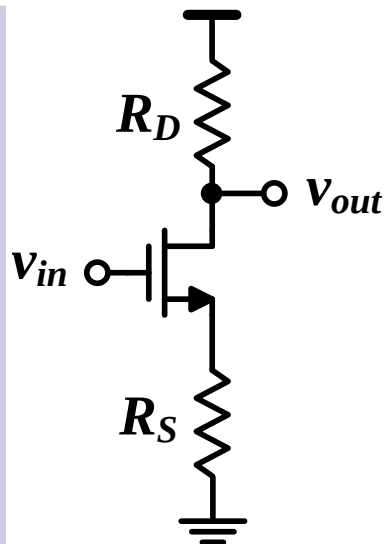
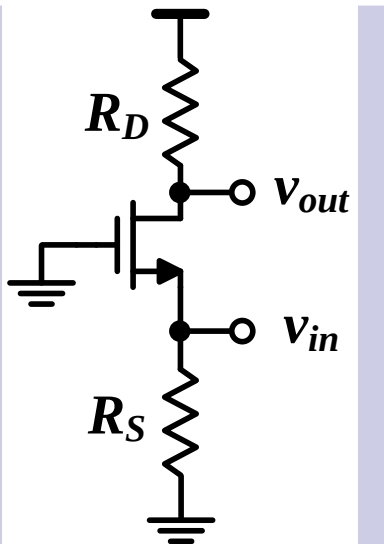
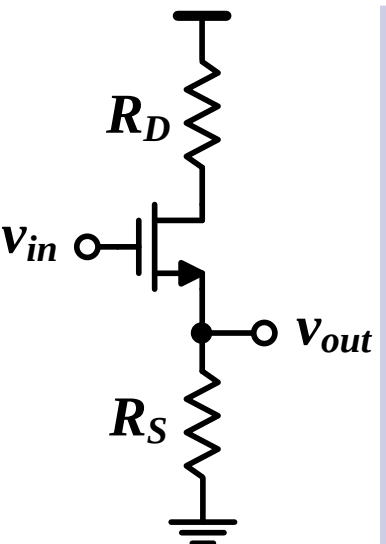


□ Divide and conquer

- Rout simplified: $v_{in} = 0$
- Gm simplified: $v_{out} = 0$
- We already need Rin/out and Gm
- We can quickly and easily get Rin/out from the shortcuts



Summary of Basic Topologies

	CS	CG	CD (SF)
			
	Voltage & current amplifier	Voltage amplifier Current buffer	Voltage buffer Current amplifier
Rin	∞	$R_S \frac{1}{g_m + g_{mb}} \left(1 + \frac{R_D}{r_o} \right)$	∞
Rout	$R_D r_o [1 + (g_m + g_{mb})R_S]$	$R_D r_o$	$R_S \frac{1}{g_m + g_{mb}} \left(1 + \frac{R_D}{r_o} \right)$
Gm	$\frac{-g_m}{1 + (g_m + g_{mb})R_S}$	$g_m + g_{mb}$	$\frac{g_m}{1 + R_D/r_o}$

Outline

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- ☐ Telescopic vs folded cascode
- ☐ Super cascode

Boosting Voltage Gain

- ❑ The max gain of a single transistor amplifier is the intrinsic gain (body effect may add).

$$A_i = g_m r_o$$

← *بشكل صغ*

- ❑ $g_m r_o$ is quite small for modern deep submicron technologies.

- May be less than 10!
- But we need high voltage gain to design an op-amp.

- ❑ How to boost the gain?

- Both CS and CG stages provide voltage gain.

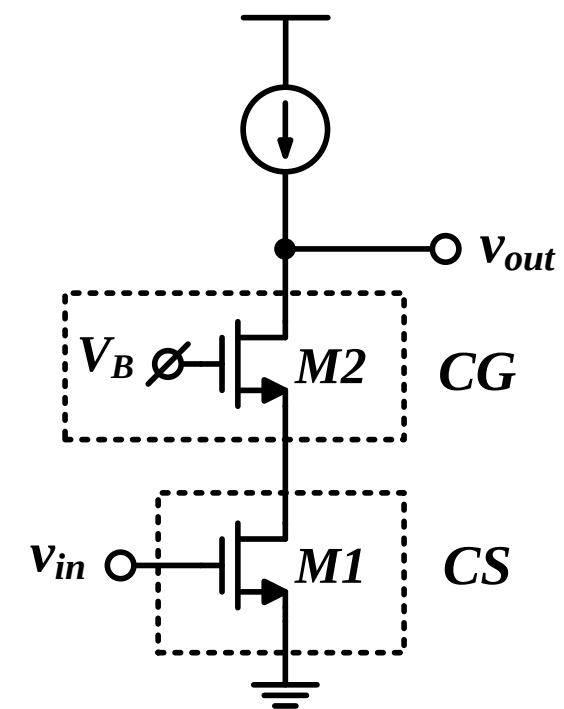
1. Use a cascade of CS stages: CS + CS + CS + ...

2. Use a cascode amplifier: CS + CG + CG + ...

- Single cascode: CS + CG
- Double cascode: CS + CG + CG
- ...

Cascode

❑ CS + CG



Cascode Gain Using GmRout

- Transconductance is always related to the input transistor (VCCS)

$$i_{out,sc} \approx -g_{m1}v_{in}$$

$$G_m = \frac{i_{out,sc}}{v_{in}} \approx -g_{m1}$$

Same Gm of CS

$$R_{out} \approx r_{o2}[1 + (g_{m2} + g_{mb2})r_{o1}]$$

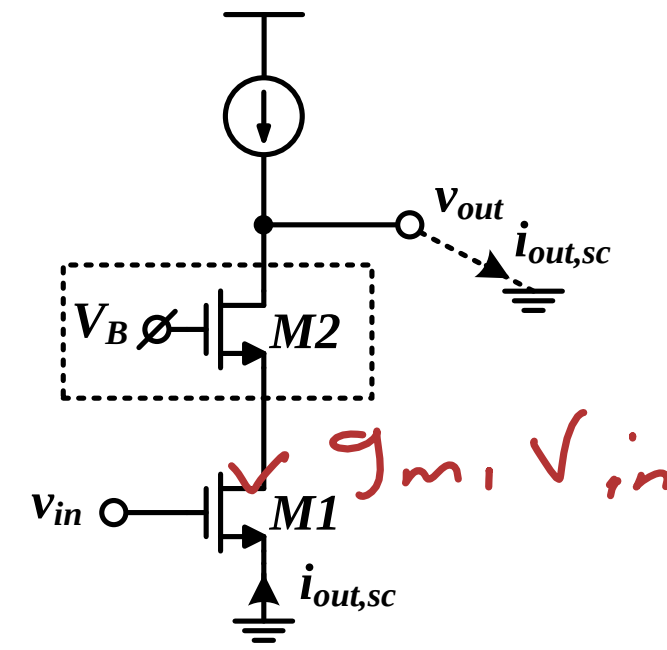
$$\approx r_{o2}(g_{m2} + g_{mb2})r_{o1}$$

Rout significantly boosted

$$A_v \approx -g_{m1}r_{o1}(g_{m2} + g_{mb2})r_{o2}$$

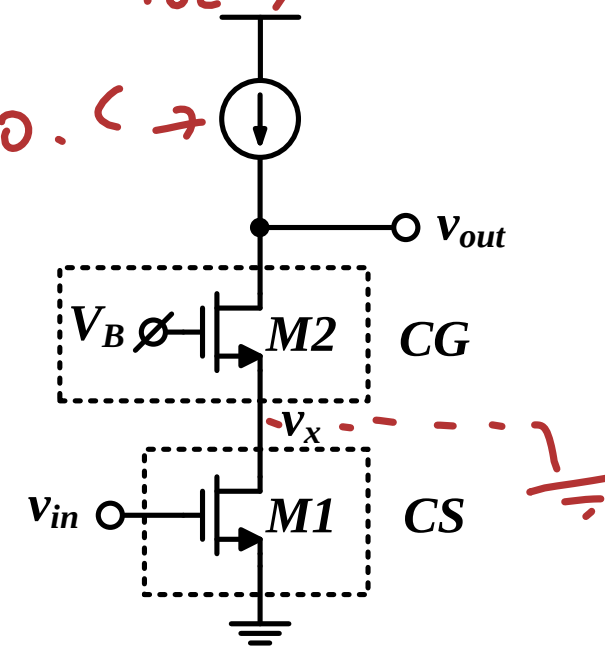
- Assume all g_m and r_o are equal and neglect body effect

$$A_v = -(g_m r_o)^2$$



Cascode as CS + CG

$$\begin{aligned}
 \text{CS: } \frac{v_x}{v_{in}} &= -g_{m1}(r_{o1}/\infty) \quad \text{Handwritten: } \frac{1}{g_m} \left(1 + \frac{\infty}{r_{o2}} \right) \\
 \text{CG: } \frac{v_{out}}{v_x} &= (g_{m2} + g_{mb2})r_{o2} \\
 A_v &= \frac{v_x}{v_{in}} \cdot \frac{v_{out}}{v_x} \\
 &\approx -g_{m1}r_{o1}(g_{m2} + g_{mb2})r_{o2} \quad \checkmark
 \end{aligned}$$



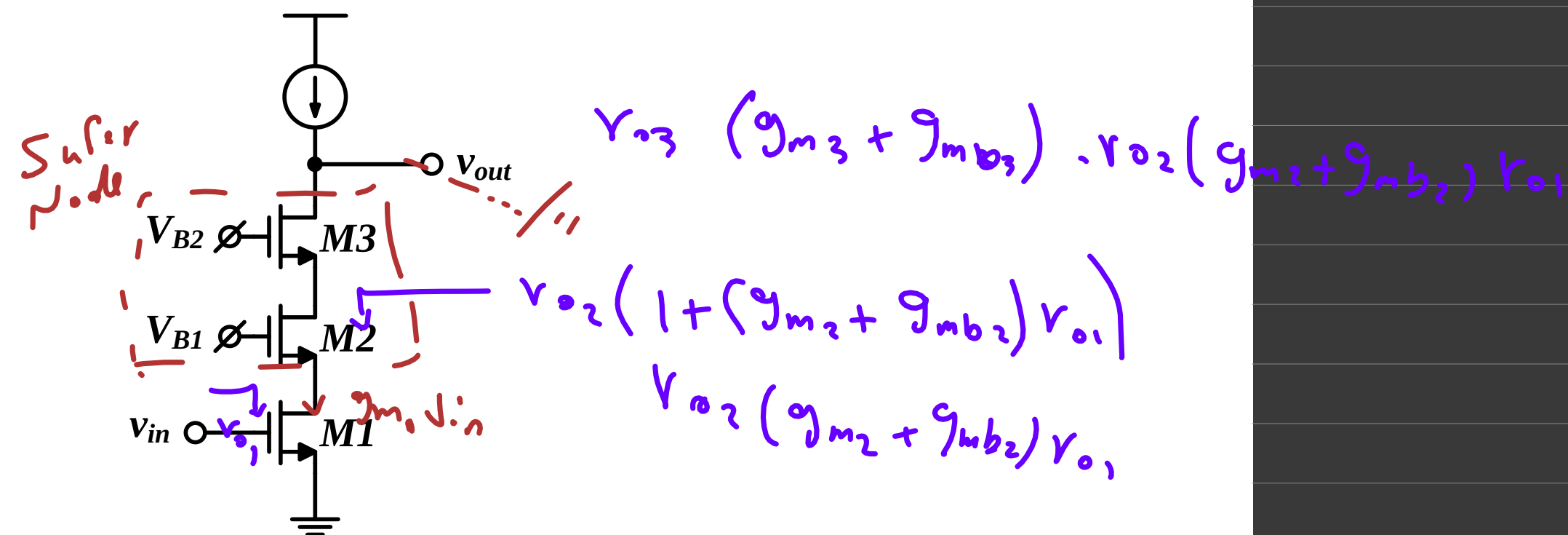
□ Assume all g_m and r_o are equal and neglect body effect

$$A_v = -(g_m r_o)^2$$

Quiz

- ❑ The circuit below shows a double cascode.
- ❑ Find the voltage gain. Assume all g_m and r_o are equal and neglect body effect.

$$G_m = -g_{m1}$$



$$A_v = - (g_m r_o)^3$$

What if R_D is small?

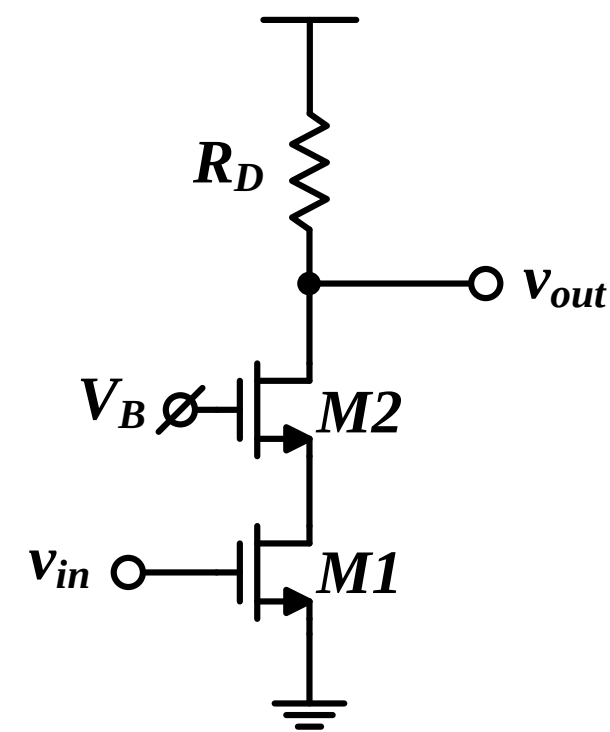
❑ Is this cascode useful?

- Not useful for boosting the gain
- But useful for boosting the BW (more about this later)

B.W extension

$$G_m = -g_{m1}$$

$$R_{out} \approx R_D$$



Cascode Load

- ❑ If you want to keep the large R_{out} , you must use cascode load

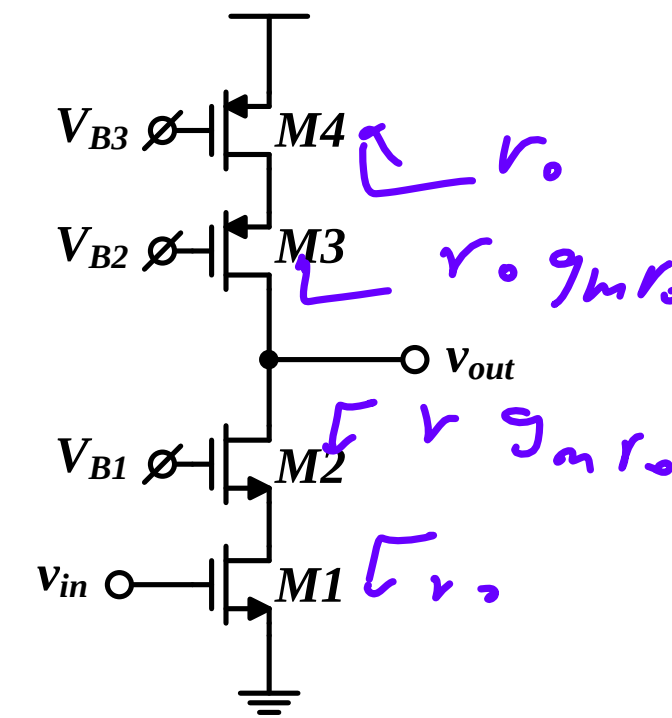
- ❑ Assume all g_m and r_o are equal and neglect body effect

$$A_v = -\frac{(g_m r_o)^2}{2}$$

- ❑ Output swing $\approx V_{DD} - 4V_{ov}$

$$Max = V_{DD} - 2V_{ov}$$

$$Min = 2V_{ov}$$



• عیب ال cascode اس بقال swing

Cascode Bias Voltage

- Keep M1 in sat

$$V_B > V_{GS2} + (V_{in,max} - V_{TH1})$$

$$V_{in,max} < V_{TH1} + V_B - V_{GS2}$$

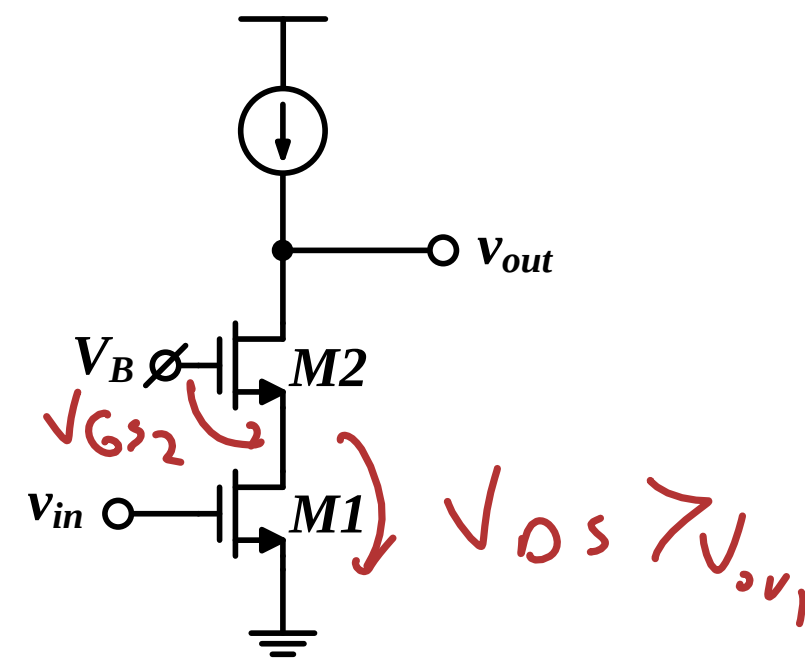
- Keep M2 in sat

$$V_B < V_{TH2} + V_{out,min}$$

$$V_{out,min} > V_B - V_{TH2}$$

- Increasing/decreasing V_B either hurts V_{in} range or V_{out} range

- Input and output ranges are coupled oppositely



برای حفظ ولتاژ تأثیر V_B و V_{in} و V_{out} یکدیگر
 می‌تواند، اگر V_B را بزرگتر کنیم، $V_{out,min}$ بزرگتر می‌شود و $V_{in,max}$ کوچکتر می‌شود

Quiz: Cascode Bias Voltage

- Assume $V_{TH} = 0.4\text{ V}$ and $V_{ov} = 0.1\text{ V}$.
- Find minimum V_B .
- If V_B is set at 0.1 V above the minimum value, find

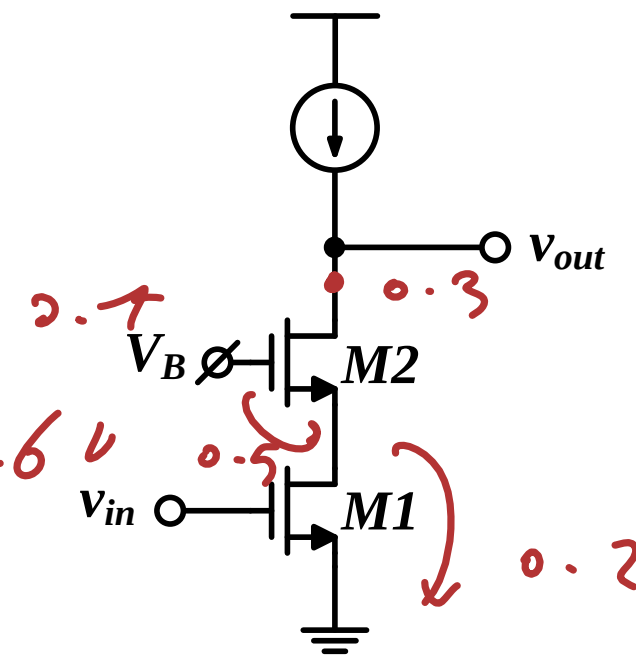
$V_{in,max}$ and $V_{out,min}$.

$$V_{B,min} = \underbrace{0.4 + 0.1}_{V_{GS2}} + \underbrace{0.1}_{V_{ov1}} = 0.6\text{ V}$$

$$V_B = 0.7$$

$$V_{in} < 0.4 + 0.2 = 0.6 = V_{in,max}$$

$$V_{out,min} = 0.3$$

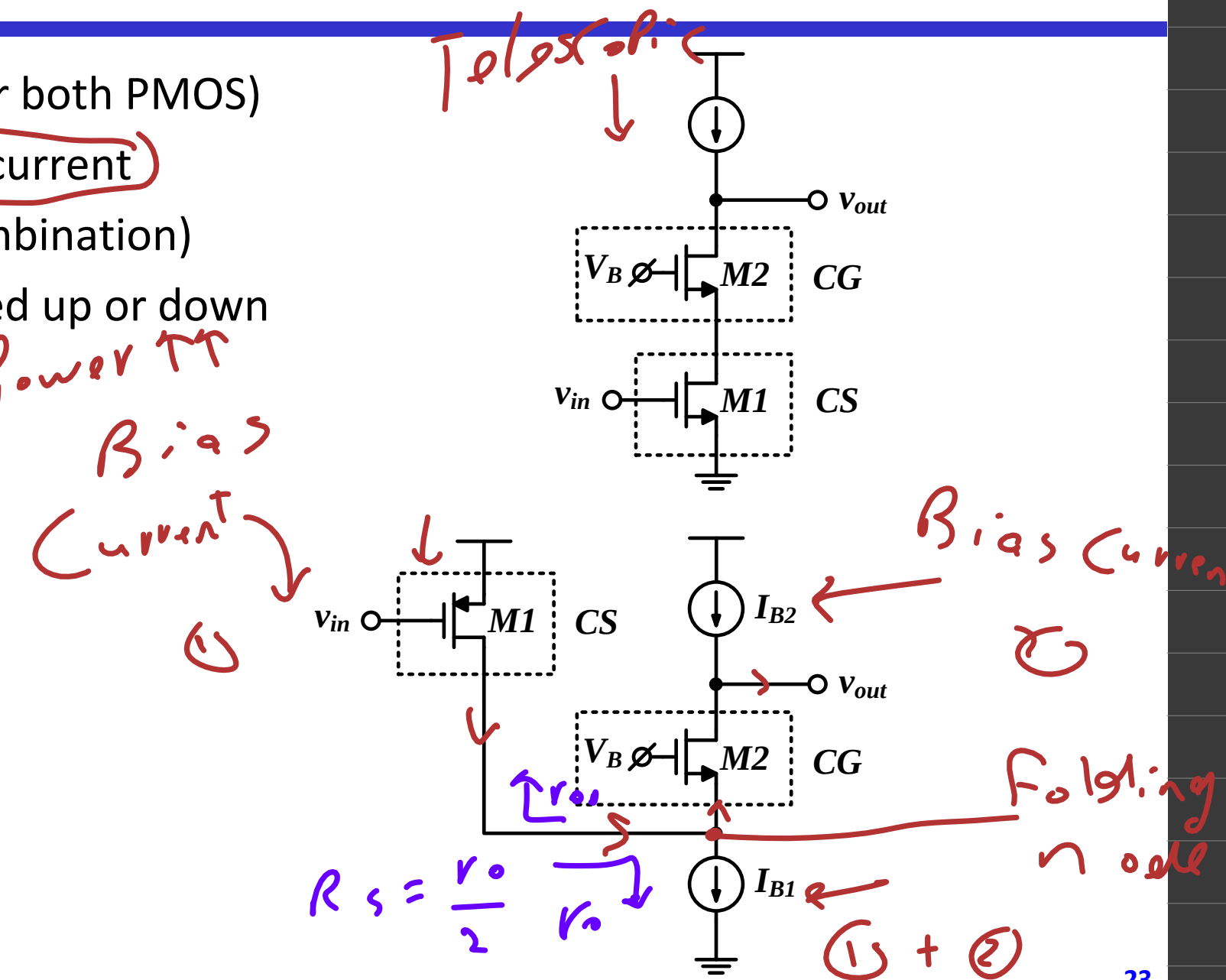


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- ☐ Super cascode

Telescopic vs Folded Cascode

- ❑ Telescopic: CS + CG (both NMOS or both PMOS)
 - Both CS and CG use same bias current
- ❑ Folded: CS + CG (NMOS-PMOS combination)
 - The small signal current is folded up or down
 - Extra bias current is needed
 - Rout is lower (due to I_{B1})
 - Why is it useful?



بیشتر جابجایی (cascode)
 VB
 کمالات بی بی
 output و input range
 0.5 r_{out}

$$R_s = \frac{r_o}{2}$$

Folded Cascode Bias Voltage

- Input and output ranges are NOT coupled oppositely

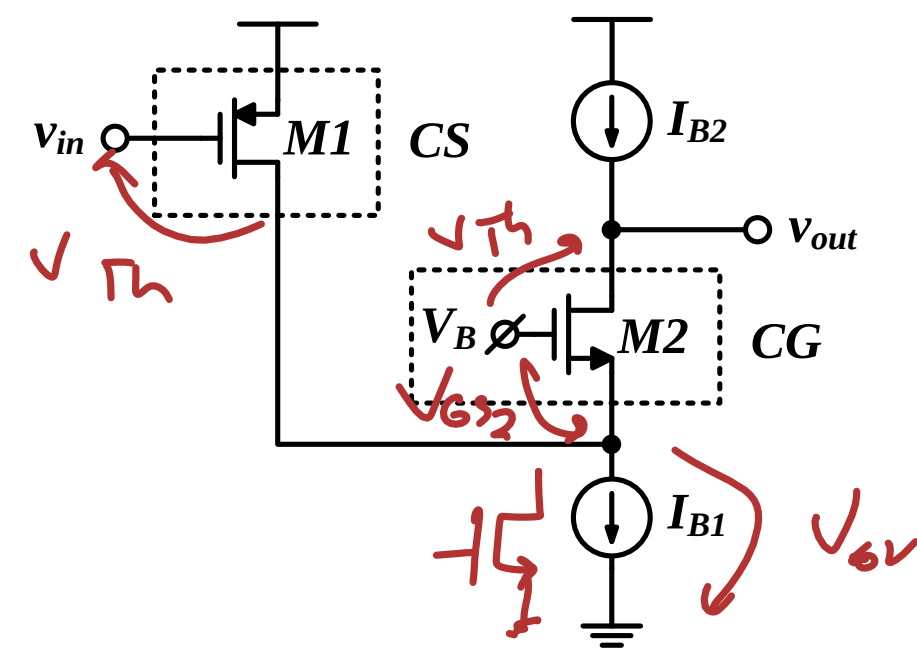
$$V_{in,min} > -|V_{TH1}| + V_B - V_{GS2}$$

$$V_{out,min} > V_B - V_{TH2}$$

- Choosing small V_B extends both V_{in} and V_{out} ranges

- Just bias I_{B1} in saturation

- More on this point when we study operational amplifier design



Quiz: Folded Cascode Bias Voltage

- Assume I_{B1} is implemented using a MOSFET, $V_{TH} = 0.4\text{ V}$, and $V_{ov} = 0.1\text{ V}$.
- Find minimum V_B .
- If V_B is set at 0.1 V above the minimum value, find $V_{in,min}$ and $V_{out,min}$.

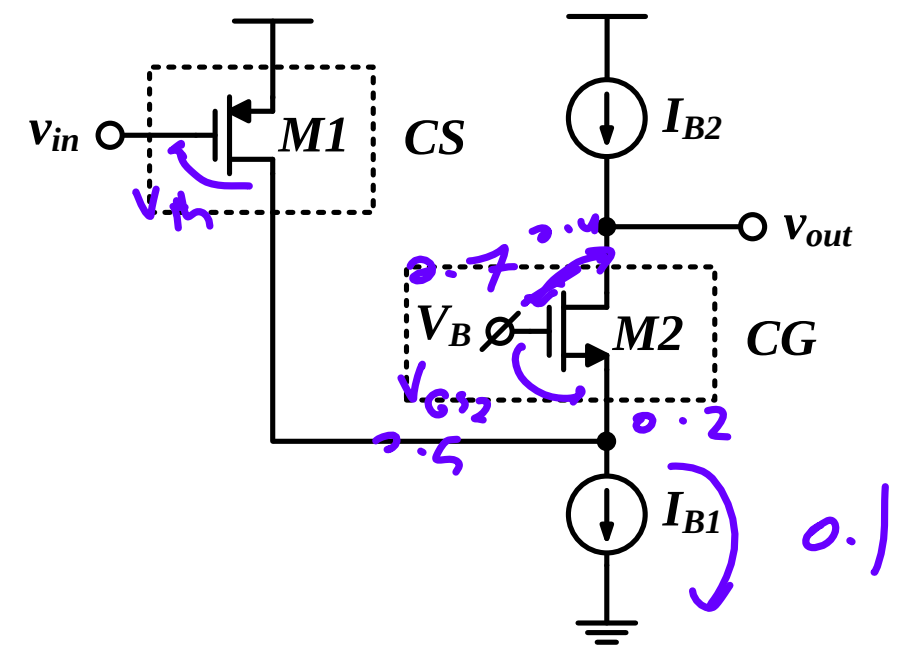
$$V_{B,min} = 0.5 + 0.1 = 0.6$$

$$V_{in,min} > -V_{th} - V_{GS2} + V_B$$

$$V_{in,min} > -0.2\text{ V}$$

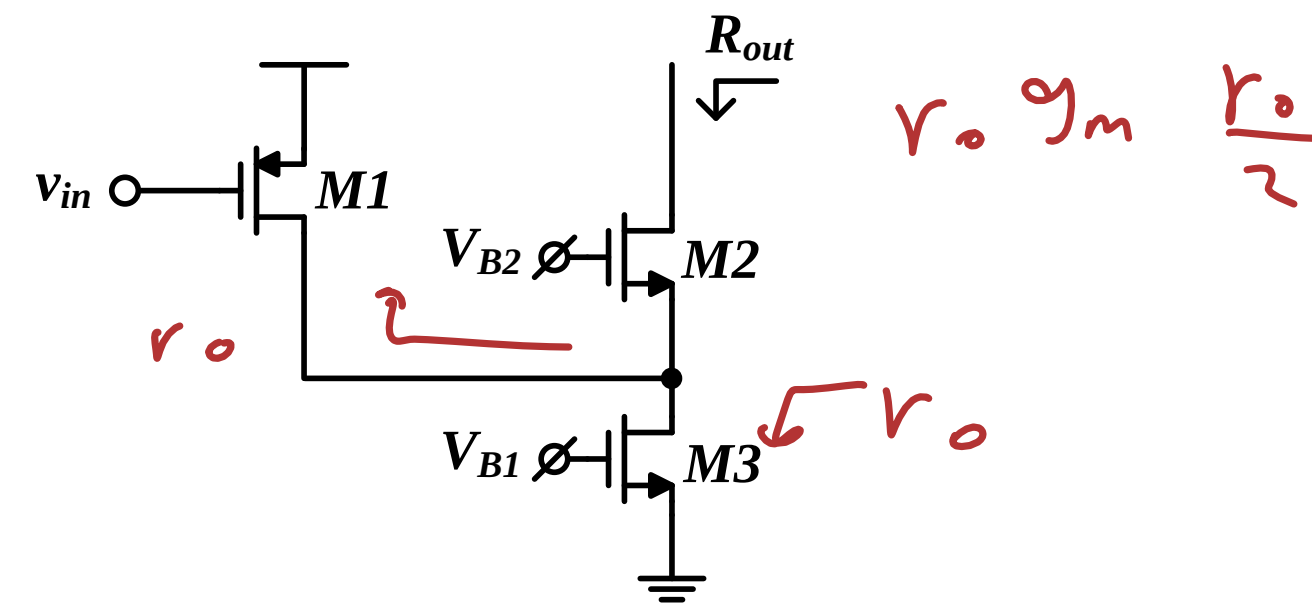
$$V_{out,min} > -V_{DS} + V_B$$

$$V_{out,min} > 0.3\text{ V}$$



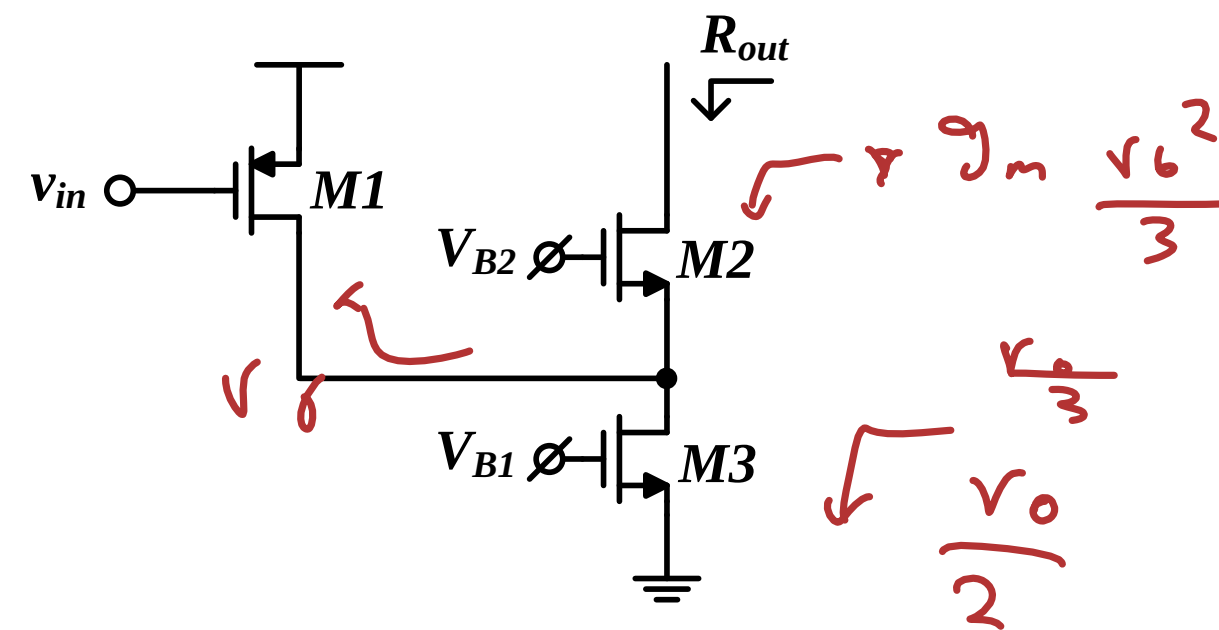
Quiz: Rout of Folded Cascode

- Assume all transistors have the same g_m and r_o , and neglect body effect. Calculate Rout.



Quiz: Rout of Folded Cascode

- Assume all transistors have the same V_{ov} and L , CS and CG have the same bias current, and neglect body effect. Calculate R_{out} in terms of g_m and r_o of M1.

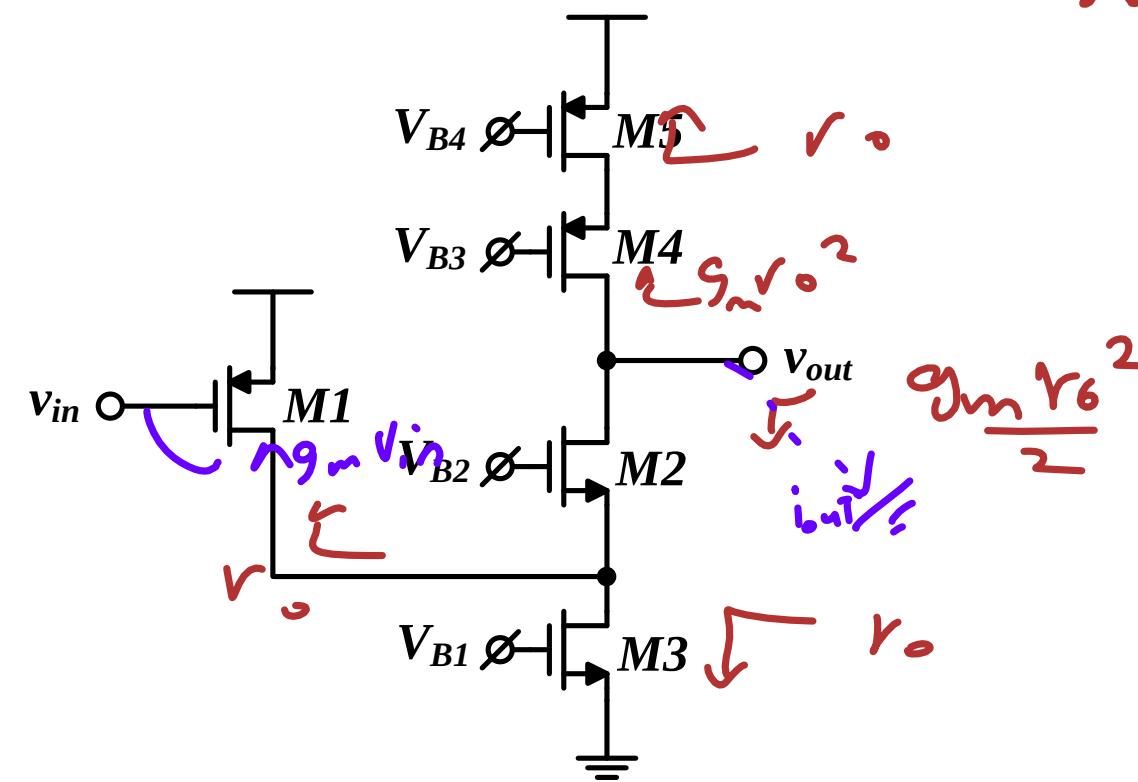


Quiz: Gain of Folded Cascode

- Calculate $A_v = G_m R_{out}$. Assume all transistors have the same g_m and r_o , and neglect body effect.

$$G_m = -g_m$$

$$R_{out} = \frac{g_m r_o^2}{3}$$



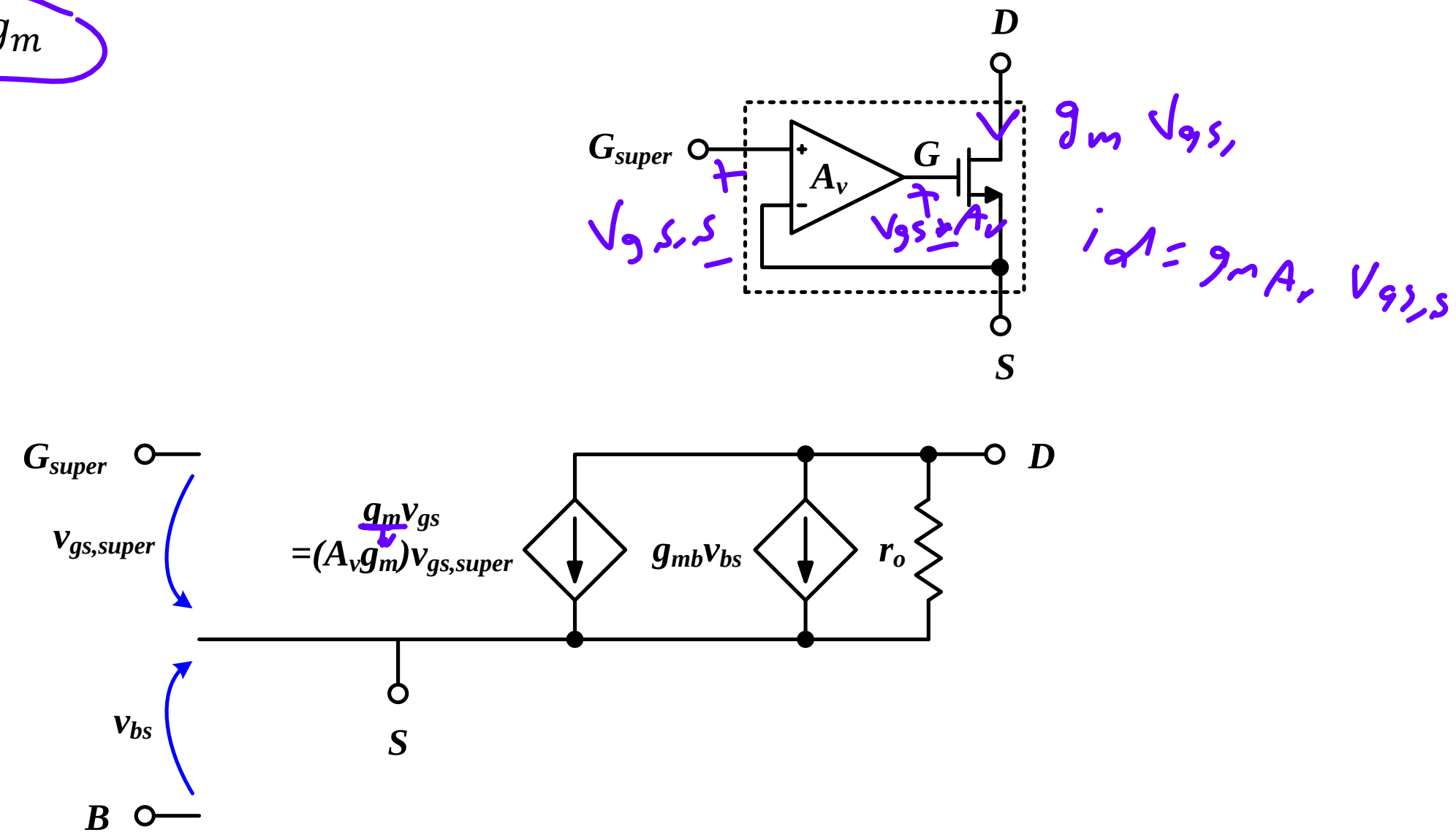
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Gain Boosting: Super Transistor

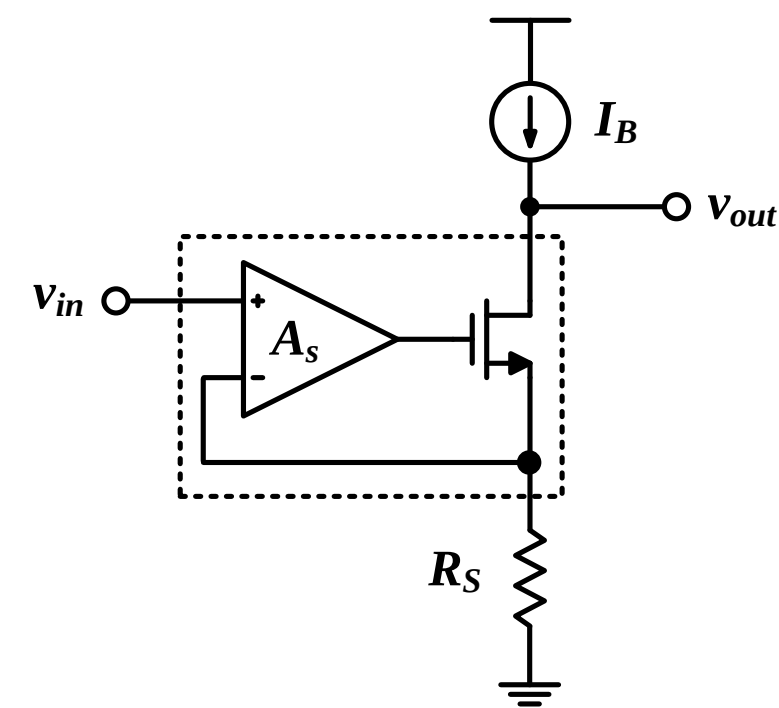
□ $g_{m,super} = A_v g_m$

□ $r_{o,super} = r_o$



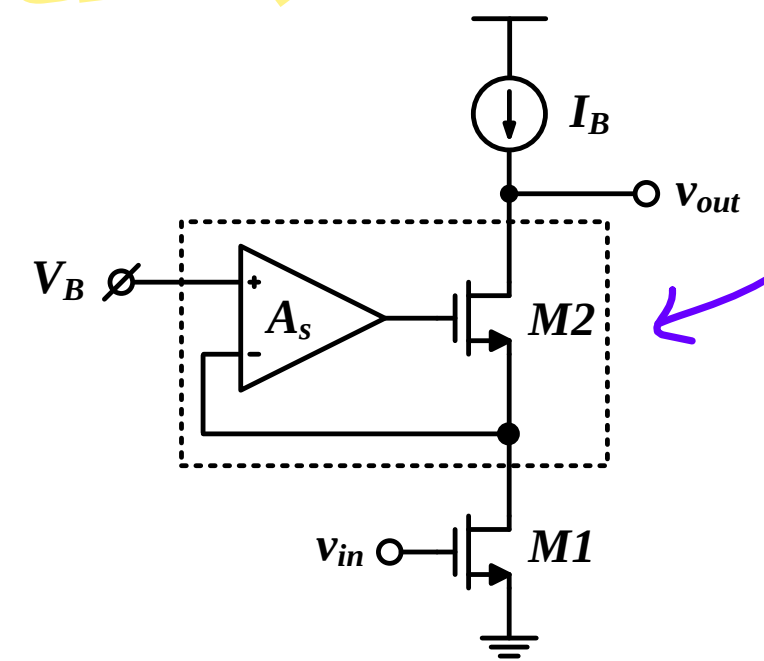
Gain Boosting: Super Transistor

- $G_m \approx \frac{g_{m,super}}{1+g_{m,super}R_S} \approx \frac{A_S g_m}{1+A_S g_m R_S} \approx \frac{1}{R_S}$
- $R_{out} \approx r_o(1 + g_{m,super}R_S) = r_o(1 + A_S g_m R_S)$ *$\approx A_S g_m R_S$*
- $A_v \approx A_S g_m r_o$



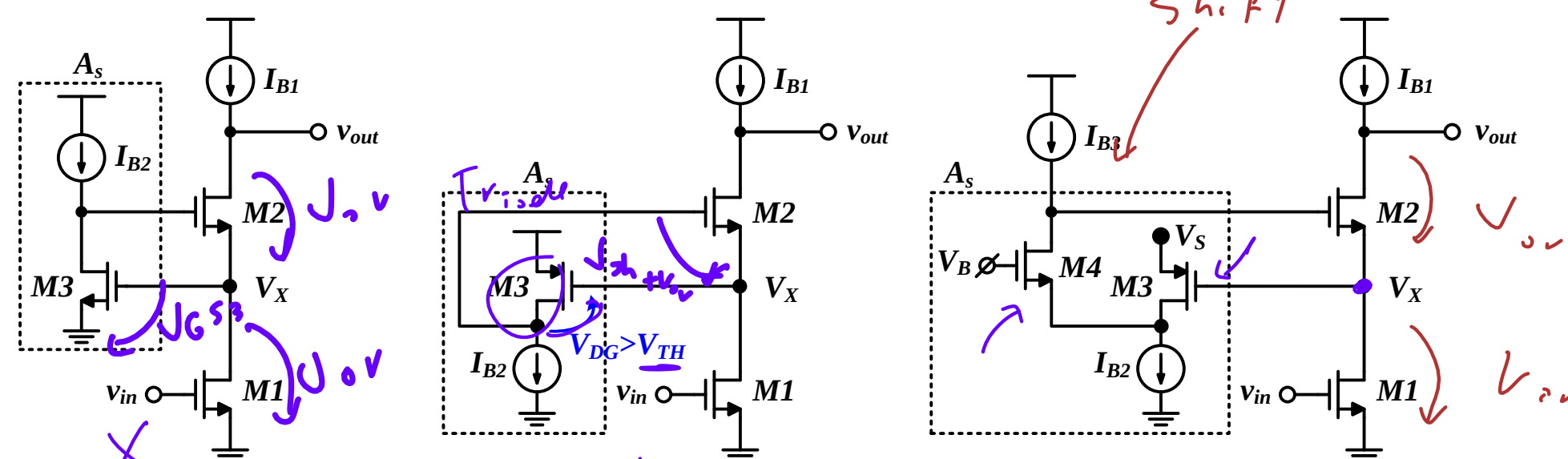
Super Cascode

- ❑ A.k.a. regulated cascode or gain boosted cascode
- ❑ $G_m \approx g_{m1}$
- ❑ $R_{out} = r_{o2}(1 + g_{m2,super}r_{o1}) = r_{o2}(1 + A_s g_{m2} r_{o1})$
- ❑ $A_v \approx A_s(g_{m1}r_{o1})(g_{m2}r_{o2})$ $(g_{m1}r_{o1})^2 A_s$
- ❑ Gain is boosted while preserving headroom (no extra stacking)
- ❑ But more power, noise, and complexity



Gain Boosting Implementation

- ❑ NMOS CS: headroom limitation
 - $V_X = V_{GS3} = V_{TH3} + V_{ov3}$ instead of V_{ov1}
- ❑ PMOS CS: M3 in triode
 - $V_{DG3} = V_{GS2} > V_{TH}$
- ❑ Folded cascode: M4 provides level shift
 - Can bias the circuit such that $V_{out,min} = V_{ov2} + V_{ov1}$



من ناحية در
headroom
آبیره

Thank you!

References

- ❑ A. Sedra and K. Smith, “Microelectronic Circuits,” Oxford University Press, 7th ed., 2015
- ❑ B. Razavi, “Fundamentals of Microelectronics,” Wiley, 2nd ed., 2014
- ❑ B. Razavi, “Design of Analog CMOS Integrated Circuits,” McGraw-Hill, 2nd ed., 2017